KAIZEN MATHS

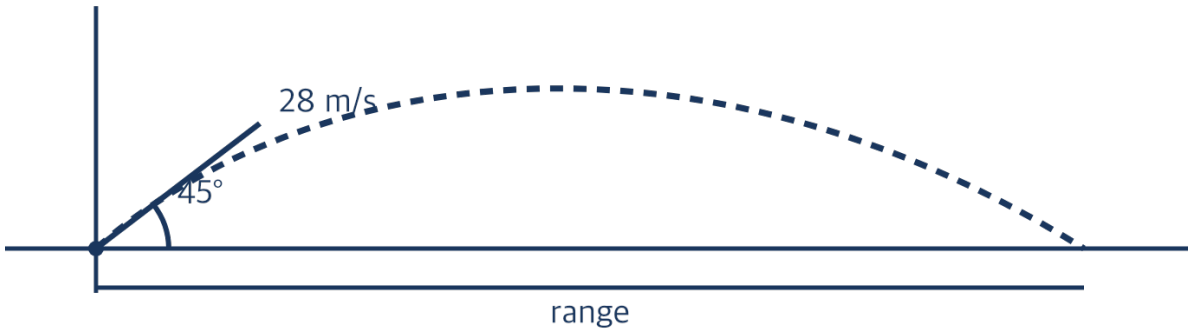
Lesson starter

Projectile Motion

Position & Range

Name _____

SA1



A projectile is launched from level ground at 28 m/s at 45°. Find its horizontal range.

MEC-5B63-L2 · SA

Show your working

KAIZEN MATHS

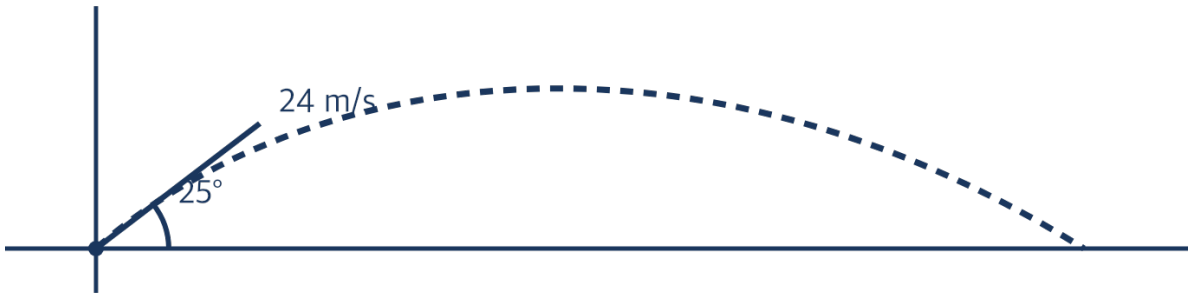
Lesson starter

Projectile Motion

Position & Range

Name _____

SB1



A projectile is launched at 24 m/s at 25°. Find its horizontal distance and height after 1 s.

MEC-5B63-L2 · SB

Show your working

KAIZEN MATHS

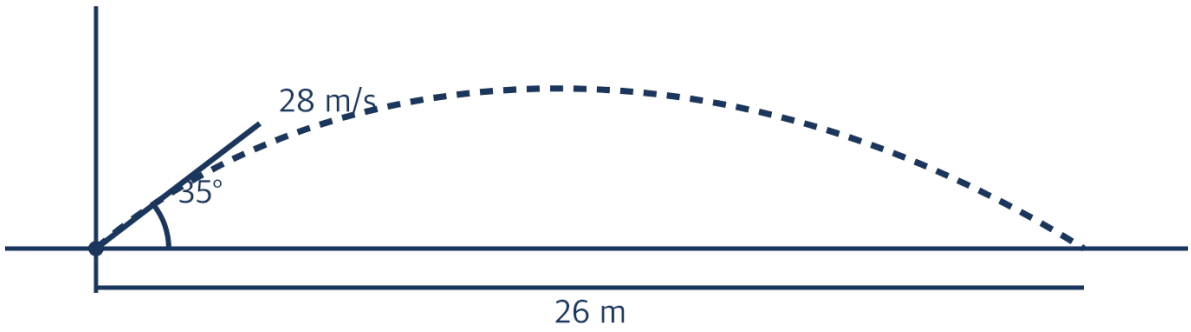
Lesson starter

Projectile Motion

Position & Range

Name _____

SC1



A projectile is launched at 28 m/s at 35°. Find its height when it has travelled 26 m horizontally.

MEC-5B63-L2 · SC

Show your working

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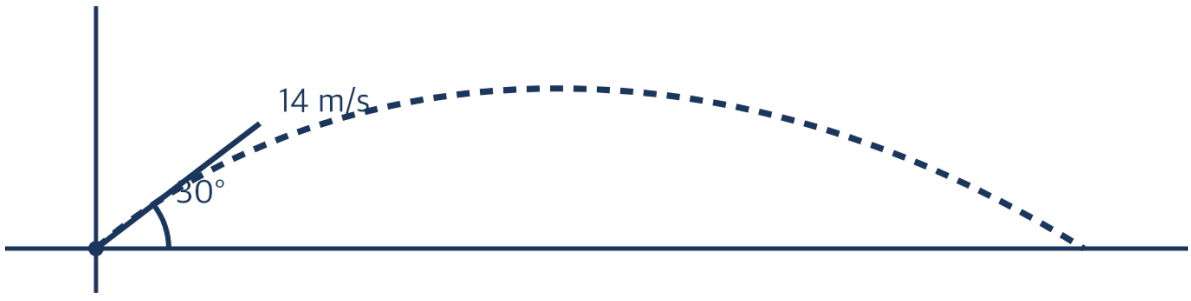
Lesson starter

Projectile Motion

Position & Range

Name _____

SD1



A projectile is launched at 14 m/s at 30°. Find its speed after 1 s.

MEC-5B63-L2 · SD

Show your working

KAIZEN MATHS

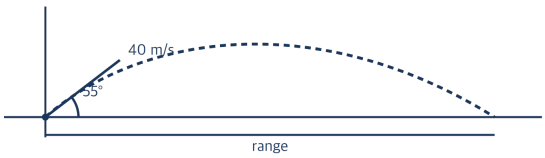
Exit ticket

Projectile Motion

Position & Range

Name _____

EA1



A projectile is launched from level ground at 40 m/s at 55°. Find its horizontal range.

Show your working

KAIZEN MATHS

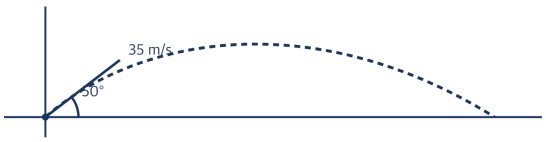
Exit ticket

Projectile Motion

Position & Range

Name _____

EB1



A projectile is launched at 35 m/s at 50°. Find its horizontal distance and height after 3 s.

Show your working

KAIZEN MATHS

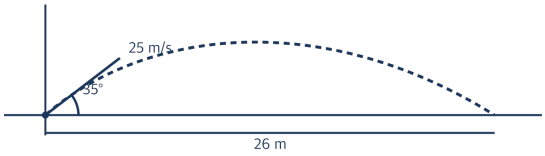
Exit ticket

Projectile Motion

Position & Range

Name _____

EC1



A projectile is launched at 25 m/s at 35°. Find its height when it has travelled 26 m horizontally.

Show your working

KAIZEN MATHS

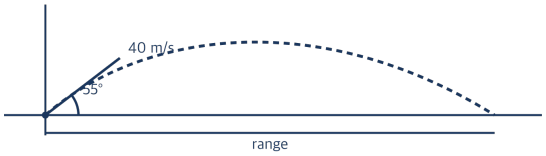
Exit ticket

Projectile Motion

Position & Range

Name _____

EA1



A projectile is launched from level ground at 40 m/s at 55°. Find its horizontal range.

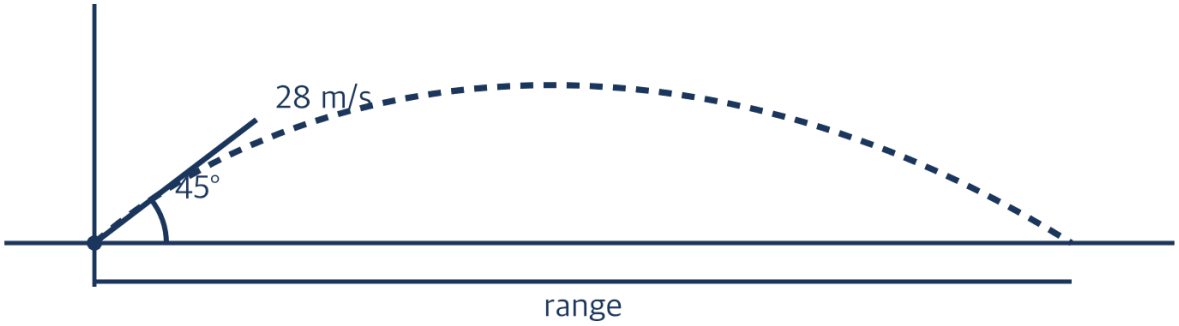
Show your working

Teacher answers & working

Projectile Motion · Position & Range

MEC-5B63-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



A projectile is launched from level ground at 28 m/s at 45°. Find its horizontal range.

Answer:

range = 80 m

Working:

Resolve the velocity into components.

$u_x = 28 \cos 45° = 19.8 \text{ m/s}$

$u_y = 28 \sin 45° = 19.8 \text{ m/s}$

Find the time of flight from the vertical motion.

$0 = 19.8t - 4.9t^2$

$t = 4.04 \text{ s}$

Use horizontal distance = horizontal speed × time.

range = 19.8 × 4.04 = 80 m

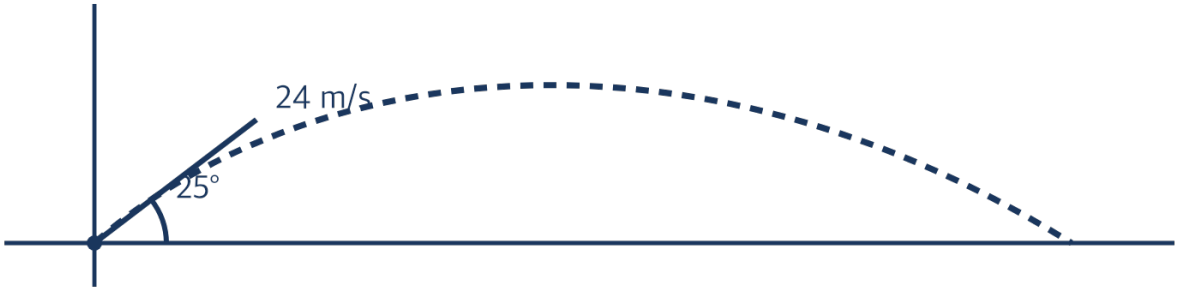
The horizontal range is 80 m.

Teacher answers & working

Projectile Motion · Position & Range

MEC-5B63-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



A projectile is launched at 24 m/s at 25°. Find its horizontal distance and height after 1 s.

Answer:

$x = 21.75$ m, $y = 5.24$ m

Working:

Resolve the initial velocity.

$u_x = 21.75$ m/s, $u_y = 10.14$ m/s

Use horizontal motion.

$x = u_x t = 21.75 \times 1 = 21.75$ m

Use vertical motion.

$y = u_y t - 4.9t^2$

$y = 10.14 \times 1 - 4.9 \times 1^2 = 5.24$ m

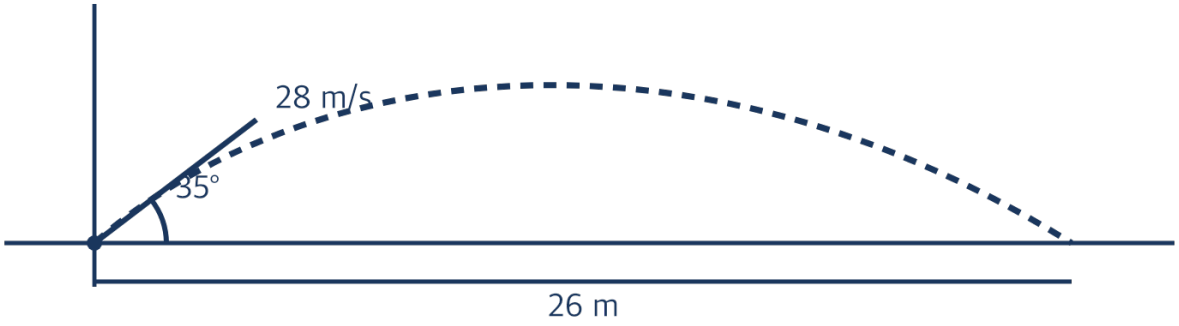
After 1 s, the projectile is 21.75 m horizontally and 5.24 m above the launch point.

Teacher answers & working

Projectile Motion · Position & Range

MEC-5B63-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



A projectile is launched at 28 m/s at 35°. Find its height when it has travelled 26 m horizontally.

Answer:

height = 11.91 m

Working:

Resolve the velocity.

$u_x = 22.94$ m/s, $u_y = 16.06$ m/s

Use horizontal motion to find the time.

$x = u_x t$, so $t = x \div u_x$

$t = 26 \div 22.94 = 1.13$ s

Substitute this time into the vertical equation.

$y = 16.06 \times 1.13 - 4.9 \times 1.13^2 = 11.91$ m

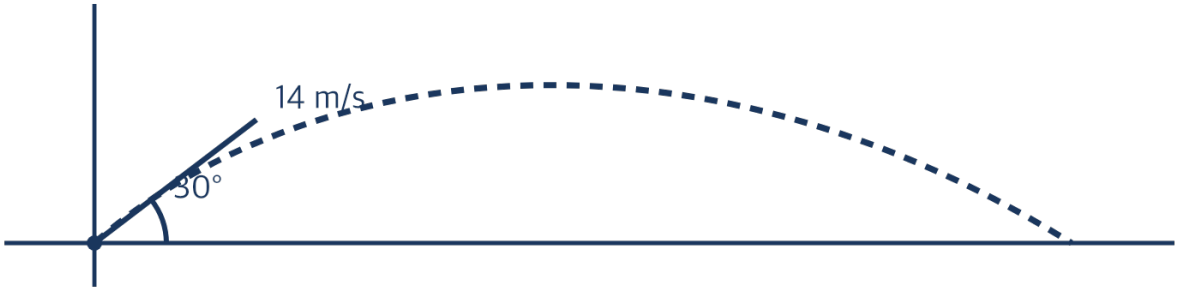
The height is 11.91 m.

Teacher answers & working

Projectile Motion · Position & Range

MEC-5B63-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



A projectile is launched at 14 m/s at 30°. Find its speed after 1 s.

Answer:

speed = 12.44 m/s

Working:

Resolve the initial velocity.

$u_x = 12.12\ \text{m/s}, u_y = 7\ \text{m/s}$

Horizontal velocity is constant.

$v_x = 12.12\ \text{m/s}$

Use vertical velocity after 1 s.

$v_y = u_y - 9.8t = 7 - 9.8 \times 1 = -2.8\ \text{m/s}$

Use Pythagoras on the velocity components.

speed = $\sqrt{(12.12^2 + -2.8^2)}$ = 12.44 m/s

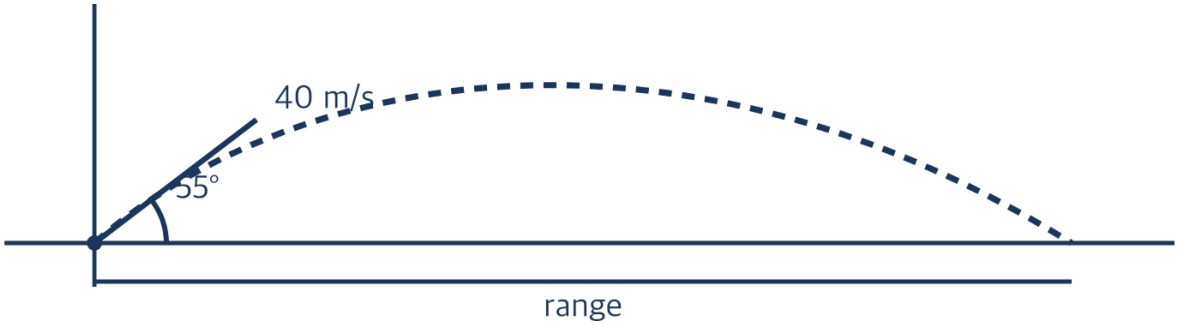
The speed after 1 s is 12.44 m/s.

Teacher answers & working

Projectile Motion · Position & Range

MEC-5B63-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



A projectile is launched from level ground at 40 m/s at 55°. Find its horizontal range.

Answer:

range = 153.42 m

Working:

Resolve the velocity into components.

$u_x = 40 \cos 55° = 22.94 \text{ m/s}$

$u_y = 40 \sin 55° = 32.77 \text{ m/s}$

Find the time of flight from the vertical motion.

$0 = 32.77t - 4.9t^2$

$t = 6.69 \text{ s}$

Use horizontal distance = horizontal speed × time.

range = 22.94 × 6.69 = 153.42 m

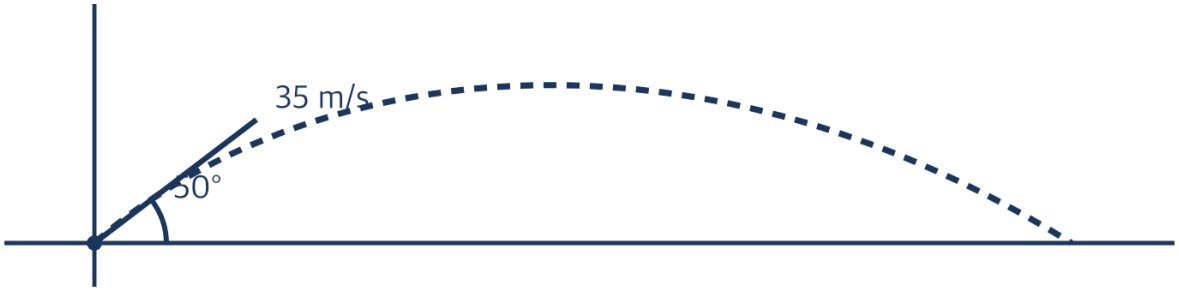
The horizontal range is 153.42 m.

Teacher answers & working

Projectile Motion · Position & Range

MEC-5B63-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



A projectile is launched at 35 m/s at 50°. Find its horizontal distance and height after 3 s.

Answer:

$x = 67.49$ m, $y = 36.33$ m

Working:

Resolve the initial velocity.

$u_x = 22.5$ m/s, $u_y = 26.81$ m/s

Use horizontal motion.

$x = u_x t = 22.5 \times 3 = 67.49$ m

Use vertical motion.

$y = u_y t - 4.9t^2$

$y = 26.81 \times 3 - 4.9 \times 3^2 = 36.33$ m

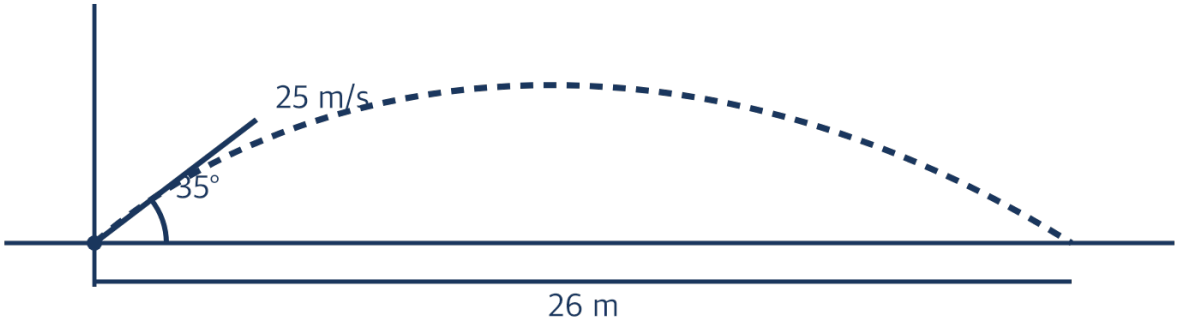
After 3 s, the projectile is 67.49 m horizontally and 36.33 m above the launch point.

Teacher answers & working

Projectile Motion · Position & Range

MEC-5B63-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



A projectile is launched at 25 m/s at 35°. Find its height when it has travelled 26 m horizontally.

Answer:

height = 10.31 m

Working:

Resolve the velocity.

$u_x = 20.48$ m/s, $u_y = 14.34$ m/s

Use horizontal motion to find the time.

$x = u_x t$, so $t = x \div u_x$

$t = 26 \div 20.48 = 1.27$ s

Substitute this time into the vertical equation.

$y = 14.34 \times 1.27 - 4.9 \times 1.27^2 = 10.31$ m

The height is 10.31 m.